



Computer Assignment 3 & 4
Mechanics of Solids, TME235

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1. Computer Assignment 3

1.1 Problem Statement:

An axisymmetric plate is free at the inner radius and supported by a joint (which allows no deformation but cannot resist any moment) at the outer radius as shown in Fig. 1.1. The plate is subjected to a vertical loading q . The plate has a thickness h , inner radius a , outer radius b , Young's modulus E , Poisson's ratio ν , and yield stress σ_y .

The objective is to compare the radial stress σ_{rr} for $z = h/2$ and the displacement $w(r)$ for $z = h/2$, using two methods:

1. Method (a): Using an axisymmetric plate model, solve for the integration constants numerically and plot the results in MATLAB.
2. Method (b): Using Abaqus to solve the general equilibrium equation for axisymmetric conditions.

Further analysis is to examine how maximum deflection and von Mises stress changes when the plate thickness h is reduced by half, for both the axisymmetric plate model and the Abaqus FE model. In Abaqus, the analysis also focuses on identifying any stress concentrations in the FE model and comparing these stress concentrations using two different mesh sizes to see how refining the mesh affects the stress distribution.

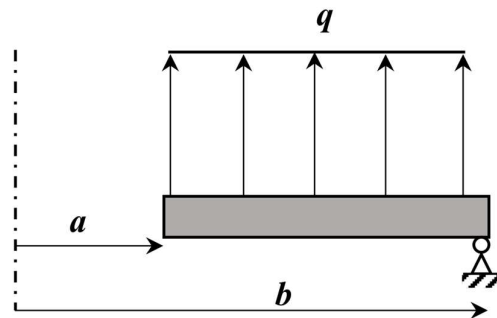


Figure 1.1

Material Properties: $q = 10$ MPa, $b = 0.2$ m, $a = 0.1$ m, $h = 0.02$ m, $E = 210$ GPa, $\nu = 0.3$ and $\sigma_y = 400$ [MPa]

1.2 Axisymmetric Plate Model Using Python:

1.2.1 Governing Equations

For Axisymmetric plate following are the governing equations which will be used to solve for deflection $w(r)$ and radial stress σ_{rr} .

$$w(r) = \int \left(\frac{1}{r} \int \left(r \int \left(\frac{1}{r} \int \frac{qr}{D} dr \right) dr \right) dr \right) dr + A_1 r^2 \ln(r/b) + A_2 r^2 + A_3 \ln(r/b) + A_4 \quad (1)$$

Here A_1, A_2, A_3 and A_4 are integration constants and D is the stiffness of the plate which is given by Eq (3).

$$\begin{bmatrix} \sigma_{rr} \\ \sigma_{\varphi\varphi} \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{bmatrix} -zw''(r) \\ -zw'(r)/r \end{bmatrix} \quad (2)$$

$$D = \frac{Eh^3}{12(1-\nu^2)} \quad (3)$$

The Boundary conditions for the plate is:

- At $r = a, M_r(r) = 0, V(r) = 0$
- At $r = b, w(r) = 0, M(r) = 0$

Here M_r is moment per unit length, and V is Shear force per unit length and their equation are given by Eqs (4) and (5) Respectively

$$\begin{bmatrix} M_r \\ M_\varphi \end{bmatrix} = D \begin{bmatrix} -w''(r) - \nu w'(r)/r \\ -w'(r)/r - \nu w''(r) \end{bmatrix} \quad (4)$$

$$V = M'_r + \frac{1}{r}(M_r - M_\varphi) \quad (5)$$

Using the above equations and boundary conditions we've computed analytical solution for axisymmetric plate as shown in the *#Snippet 1* below:

#Snippet 1:

```
import numpy as np
import matplotlib.pyplot as plt
from sympy import *

r,A1,A2,A3,A4=symbols('r,A1,A2,A3,A4',real=True)

b=0.2
a=0.1
h=0.02
E=210e9
```

```

v=0.3
D=E*h**3/(12*(1-v**2))
print(D)
q= 10e6

w=integrate(1/r*integrate(r*integrate(1/r*integrate(q*r/D,r),r),r),r)
+A1*r**2*log(r/b) + A2*r**2 + A3*log(r/b) + A4

wprime=diff(w,r)

Mr=D*(-diff(wprime,r)-v/r*wprime ) #radial bending moment
Mphi=D*(-1/r*wprime -v*diff(wprime,r))#circumf bending moment

V=diff(Mr,r)+1/r*(Mr-Mphi)

eqns = [
    w.subs(r,b),
    Mr.subs(r,b),
    Mr.subs(r,a),
    V.subs(r,a)] #bc's

unknowns=(A1,A2,A3,A4)

sol=solve(eqns,unknowns)

```

Von-Misses is calculated using the following equation:

$$\sigma_{eM} = \sqrt{\frac{3}{2} \sigma_{ij}^{dev} \sigma_{ij}^{dev}} \quad (6)$$

1.2.2 Results

Using *#Snippet 1* we found the deflection $w(r)$ at $z = 0$ to be 6.49mm, and variation of $w(r)$ along the radial length is given by Fig 1.2.

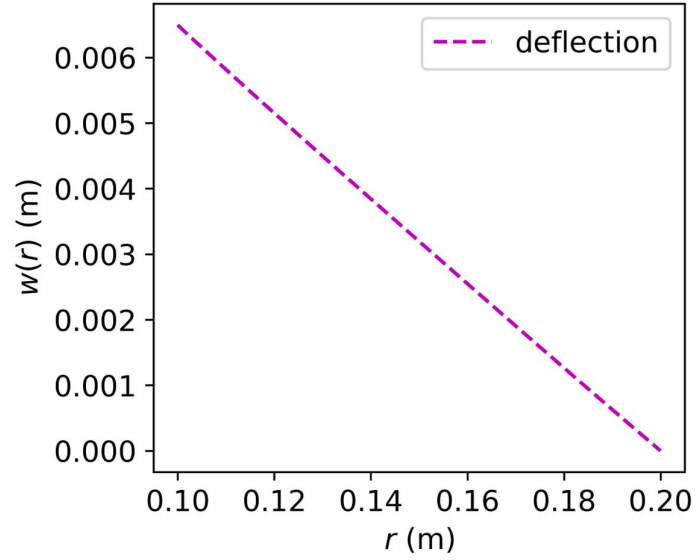


Figure 1.2 Deflection of the plate along the radial length

Also, we found radial stress σ_{rr} along the radial length of the plate at $z = h/2$ and it is shown by Fig 1.3

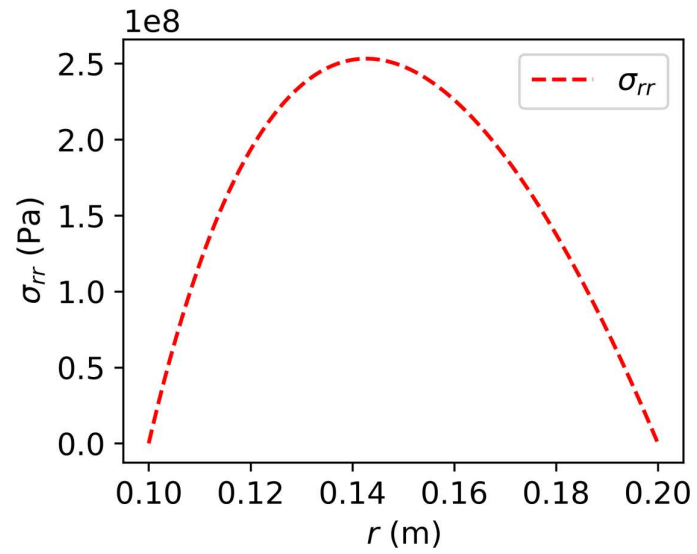


Figure 1.3 σ_{rr} of the plate along the radial length

1.3 Axisymmetric Plate Model using Abaqus :

The model shown in Fig 1.1 was modelled using Abaqus and further investigations were performed. The model was meshed using quad element with element size of 1mm as shown in Fig. 1.4.

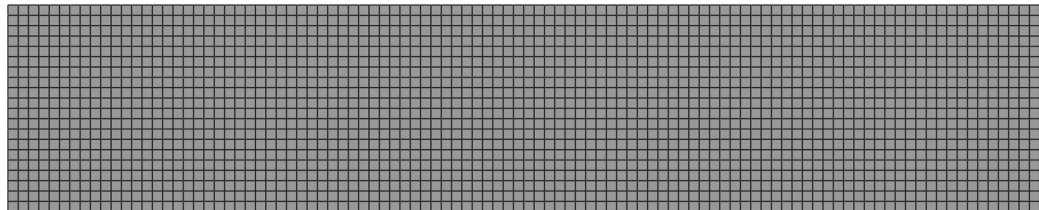


Figure 1.4 Discretization of axisymmetric plate

The boundary conditions are applied on the model as shown in Fig. 1.5, with pinned joint at the right end, and surface traction (q) on top surface of the model.

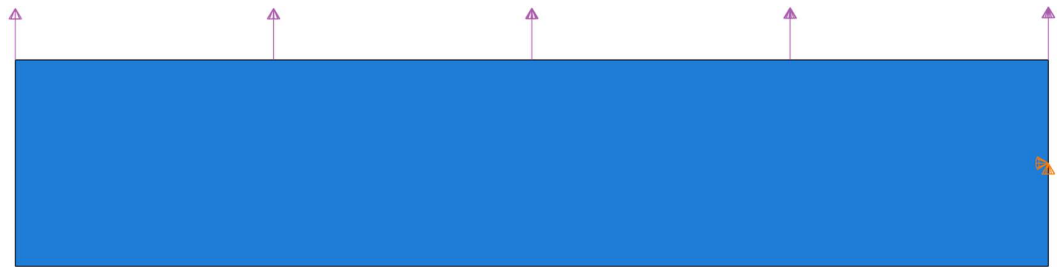


Figure 1.5 Boundary conditions for axisymmetric plate

Further the deflection of beam and radial stress are found as shown in Fig. 1.6 and Fig. 1.7 respectively.

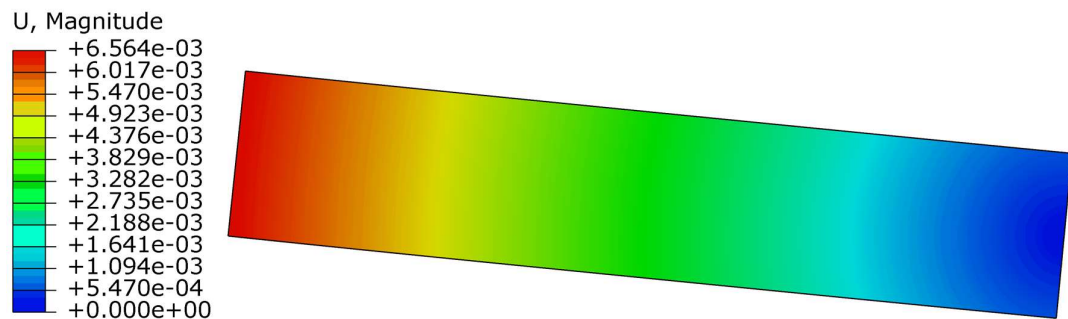


Figure 1.6 Deflection Contour plot for axisymmetric plate

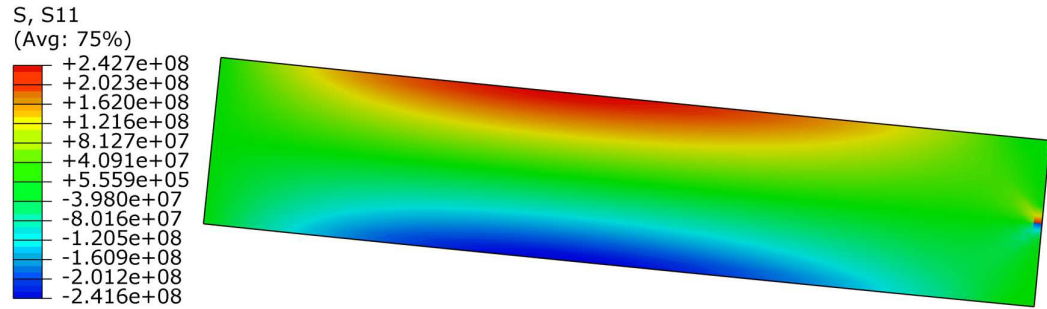


Figure 1.7 Radial Stress Contour plot for axisymmetric plate

1.4 Comparison of two Methods:

In this section we will compare the results we obtained using analytical method and FEM results.

It can be observed from the Fig. 1.8 that the solution we got for analytical and FEM are quite similar for the axisymmetric plate. So, the radial displacement in the plate is 6.49 mm and 6.56 mm for Analytical method and FEM respectively.

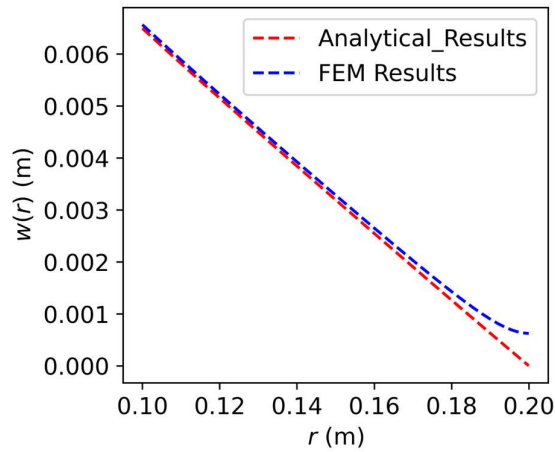


Figure 1.8 Deflection of the plate along the radial length for analytical and FEM method

Similarly, we can be variation of σ_{rr} at $z = h/2$ for both Analytical method and FEM. From the Fig. 1.9 it can be observed that the results we got for analytical and FEM are quite similar for the axisymmetric plate. So, the maximum radial displacement in the plate is 253.2 Mpa and 242.7 Mpa for Analytical method and FEM respectively.

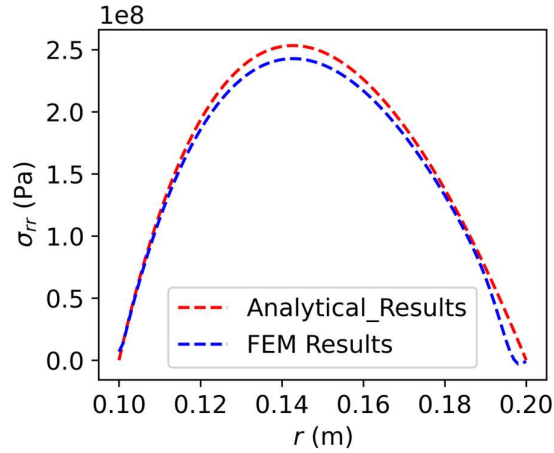


Figure 1.9 Radial Stress of the plate at $z = h/2$ for analytical method and FEM

1.5 Impact of Reducing Plate Thickness by Half on Axisymmetric Plate Results:

After reducing the plate thickness by a factor of 2, it is observed that the deflection $w(r)$ is increased from 6.49mm to 51.9 mm, and when Von-Misses stress is calculated it is observed that it has increased from 1.45 *GPa* to 5.7 *GPa*. This change can be observed in Fig 1.10 where it shows how deflection $w(r)$ as increased along the radial length, similarly from Fig 1.11 we can observe changes of Von-Misses Stress along the radial.

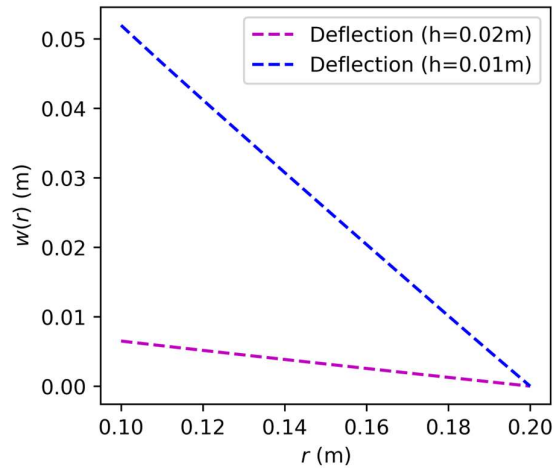


Figure 1.10 Deflection of the plate along the radial length for different thickness of the plate

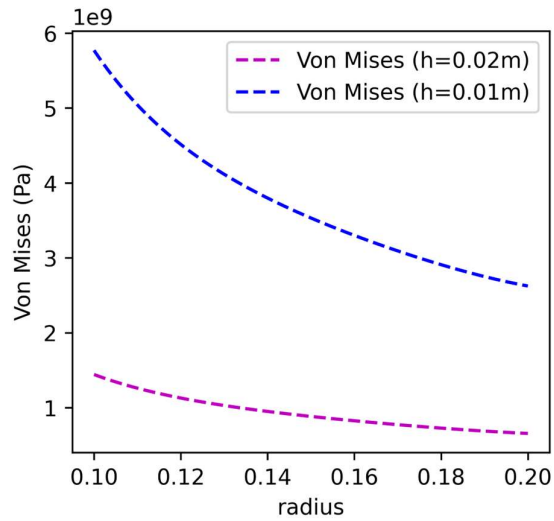


Figure 1.11 Radial Stress of the plate at $z = h/2$ for different thickness of the plate

1.6 Effect of mesh refinement on Von-Misses Stress:

The model is further refined to check its variation on Von-Misses Stress, Fig. 1.12 and Fig. 1.13 shows discretization of the model for 1mm and 0.1 mm respectively.

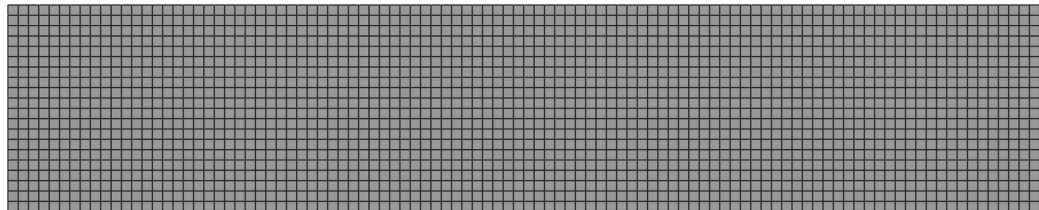


Figure 1.12 Mesh with element size = 1mm

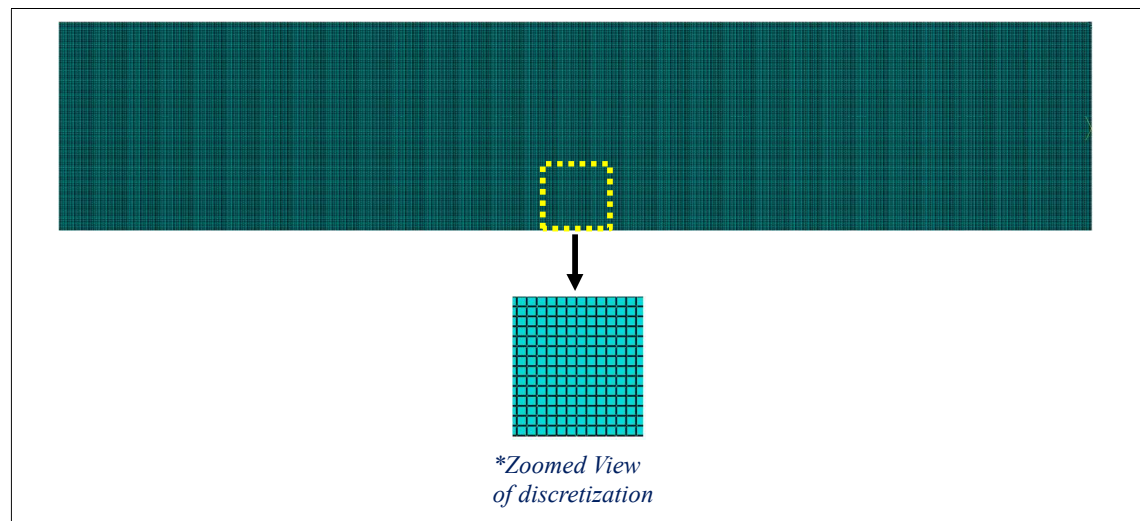


Figure 1.13 Mesh with element size = 0.1mm

From Fig. 1.14 and Fig. 1.15 it can be observed that the von misses stress has been increased from 5.2 *GPa* to 5.7 *GPa* on decreasing element size from 1mm to 0.1mm. This is because von Mises stress is mesh-dependent. A coarse mesh cannot capture the steep gradient well, and the stress is averaged over a larger area. With mesh refinement, the software better approximates the actual stress distribution, capturing higher stress values that were missed previously.

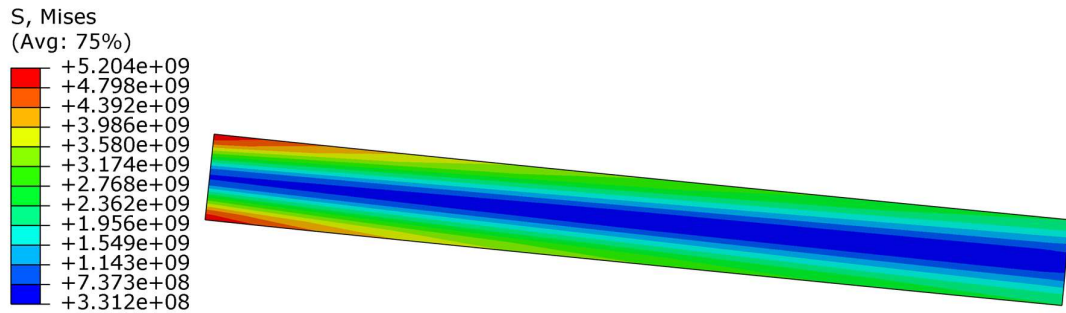


Figure 1.14 Von-Misses stress for element size 1mm

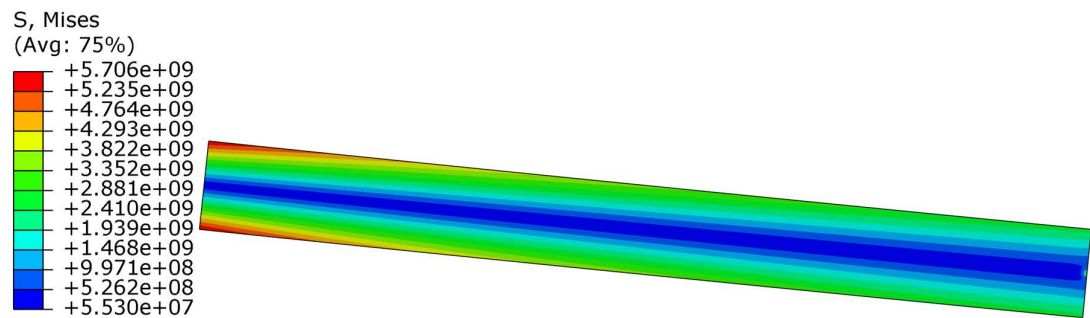


Figure 1.15 Von-Misses stress for element size 0.1mm

2. Computer Assignment 4:

2.1 Problem Statement:

A steel beam and a rubber support are attached to each other with thickness h into the plane. Here plane stress condition is assumed with steel beam being modelled as isotropic linear elastic while rubber is modelled by the compressible neo-Hooke model.

Material data is as given as below:

$$L_{steel} = 1 \text{ m}, h = 30 \text{ mm}, E_{steel} = 210 \cdot 10^3 \text{ Mpa}, \nu_{steel} = 0.3, E_{rubber} = 20 \text{ Mpa}, \nu_{rubber} = 0.45$$

2.2 True Stress v/s Stretch:

In this section we will compute the mechanical behaviour of the rubber assuming an incompressible behaviour of the rubber with stretch varying from $0.1 \leq \lambda \leq 1.9$ of a unidirectional loading.

As rubber is modelled under neo-Hooke's law, we use below formula to compute true stress in the rubber

$$\sigma = G (\lambda^2 - 1)$$

Where G is the shear modulus which is calculated using $G = E/2(1 + \nu)$ formula. Using these formulas True stress v/s Stretch has been plotted for rubber as shown below.

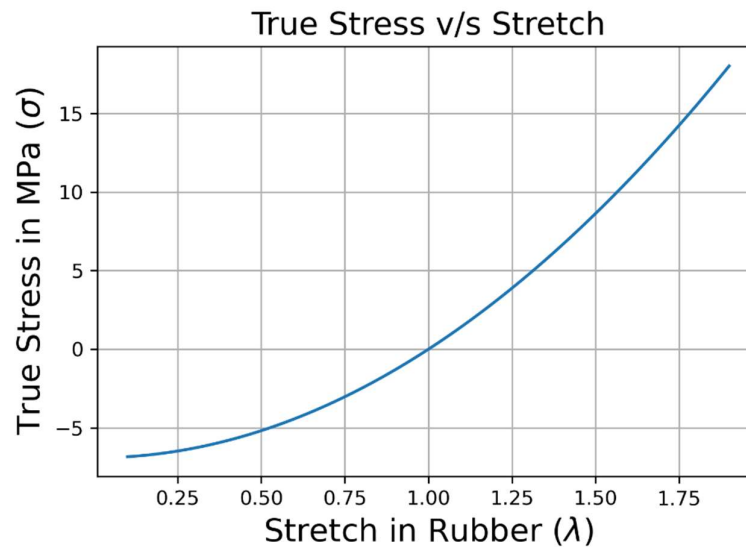


Fig 2.1. True Stress v/s Stretch for rubber

We can see that as rubber exhibits non-linear stress-strain relation, we get a curved plot rather than straight plot as we get for linear elastic relations. In the initial region we can see that stress increases slowly compared to stretch but as the stretch increases after a certain point the stress begins to increase rapidly due to strain hardening. This graph also showcases the rubber's ability to undergo large deformation (in this case 190% as

final λ value is 1.9) without permanent damage and returning to original shape when unloaded.

2.3 Vertical Deflection along Mid-Surface:

In this section we will compute deflection for the beam for 2 conditions:

- When load P is 10 kN in upward direction i.e. $P = 10 \text{ kN}$
- When load P is 10 kN in downward direction i.e. $P = -10 \text{ kN}$

This deflection is been computed at the mid surface of the beam in both conditions and while plotting the graph absolute value of deflection has been used since deflection would be negative when load is in downward direction. Below is the graph for deflection in beam for both the conditions

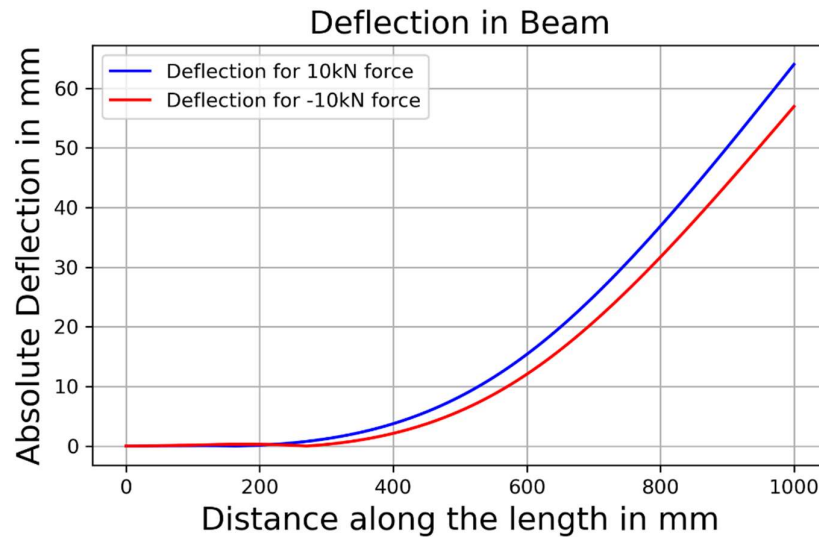


Fig 2.2. Deflection in Beam

When the force is in upward direction the rubber is under tension and it is in compression when the force is in downward direction. From the results we can see that the deflection in beam in downward direction and upward direction is around 56.95 mm and 64.02 mm respectively. This indicates that the rubber exhibits greater resistance to compression forces compared to tensile forces.

2.4 Strain Assumption:

In this assignment we've assumed linear elastic material for the steel beam and hyper elastic material (under neo-Hooke's law) for the rubber. So, basically, we are assuming small strain model for the steel beam and large strain model for rubber. And based on the contour plots of first and third principle strain below we can say that the assumption is justifiable.

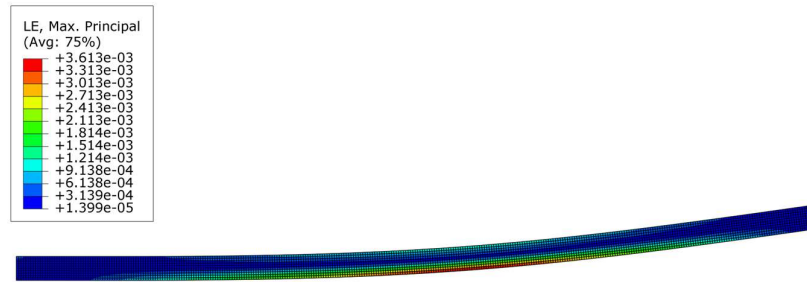


Fig 2.3 (a) First Principle strain in Beam

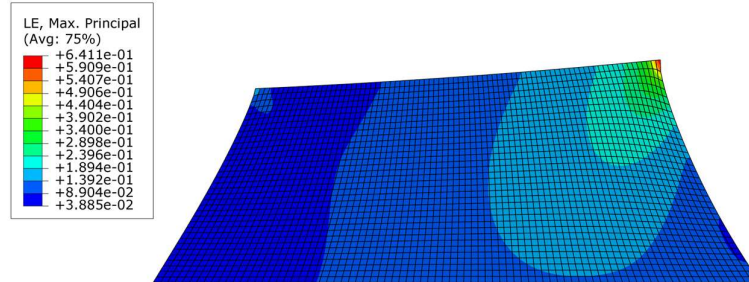


Fig 2.3 (b) First Principle strain in Rubber

Fig 2.3. First Principle Strain

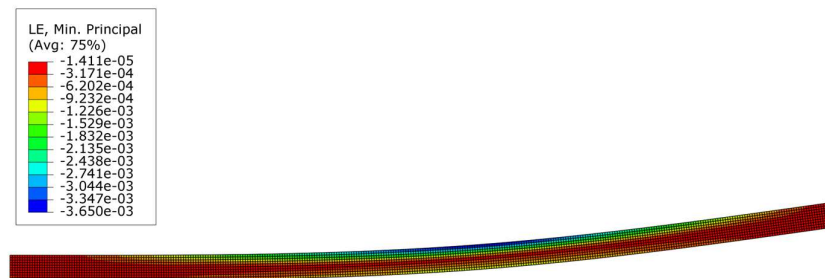


Fig 2.4 (a) Third Principle strain in Beam

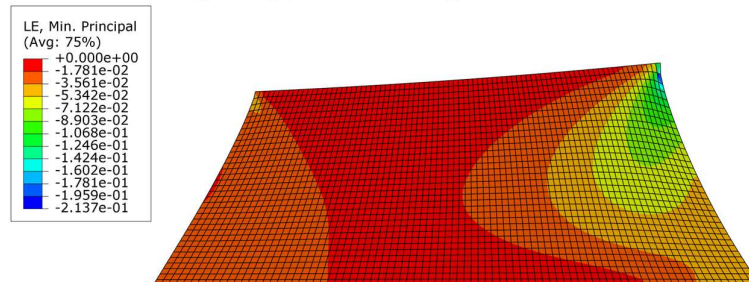


Fig 2.4 (b) Third Principle strain in Rubber

Fig 2.4. Third Principle Strain

2.5 Euler-Bernoulli Beam Model w/o rubber support:

In this section we used Euler-Bernoulli Beam Model for computing deflection in the beam without the rubber support and then the result was compared with the deflection in beam with rubber support. As we can see in the graphs below, the deflection in beam when subjected to load 10kN in upward direction and 10kN in downward direction we get a deflection of about 235.16 mm and -235.16mm respectively. We can see that the deflection values of the beam under similar loading conditions with rubber support is much smaller than the deflection values without rubber support because the rubber provides additional resistance, effectively limiting the beam's deflection.

Deflection comparison with and without rubber support for $P = 10\text{kN}$

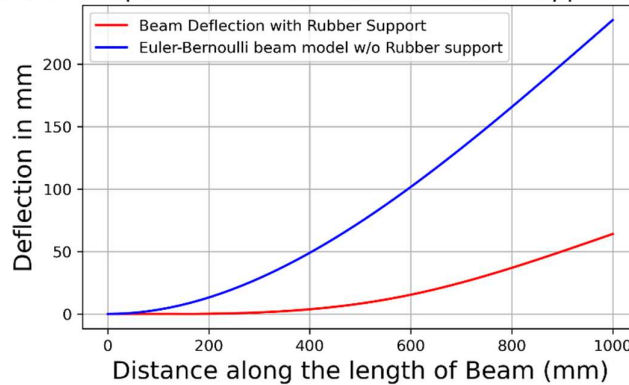


Fig 2.5 (a) Deflection comparison for $P = 10\text{kN}$

Deflection comparison with and without rubber support for $P = -10\text{kN}$

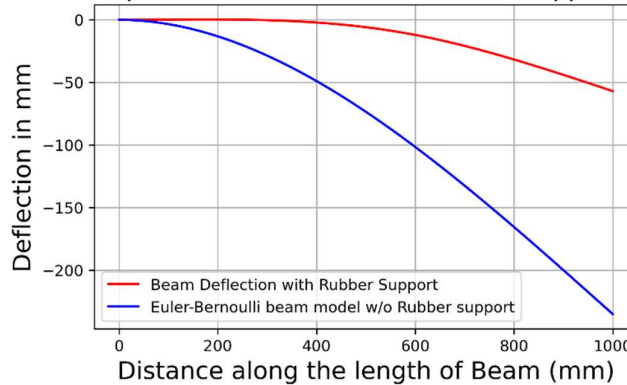


Fig 2.5 (b) Deflection comparison for $P = -10\text{kN}$

Fig 2.5. Deflection Comparison